12.679

EE25BTECH11042 - Nipun Dasari

Question:
If the area of a triangle with the vertices (k, 0), (2, 0) and (0, -2) is 2 square units, find the value of k.
Solution:

The vertices are $\mathbf{A}(k, 0)$, $\mathbf{B}(2, 0)$ and $\mathbf{C}(0, -2)$.

$$\mathbf{A} = \begin{pmatrix} k \\ 0 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 \\ -2 \end{pmatrix} \tag{0.1}$$

The side vectors are:

$$\mathbf{a} = \mathbf{B} - \mathbf{C} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \mathbf{A} - \mathbf{C} = \begin{pmatrix} k \\ 2 \end{pmatrix} \tag{0.2}$$

Area is given by $\Delta = \frac{1}{2} \|\mathbf{b} \times \mathbf{a}\|$. Since $\Delta = 2$, we have $\|\mathbf{b} \times \mathbf{a}\| = 4$.
The cross product formula is:

$$\mathbf{A} \times \mathbf{B} = \begin{pmatrix} |\mathbf{A}_{23} & \mathbf{B}_{23}| \\ |\mathbf{A}_{31} & \mathbf{B}_{31}| \\ |\mathbf{A}_{12} & \mathbf{B}_{12}| \end{pmatrix} \tag{0.3}$$

The vectors in 3-D space are $\mathbf{b} = \begin{pmatrix} k \\ 2 \\ 0 \end{pmatrix}$ and $\mathbf{a} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$.

Calculating each component:

$$|\mathbf{b}_{23} \;\; \mathbf{a}_{23}| = \begin{vmatrix} 2 & 2 \\ 0 & 0 \end{vmatrix} = 0 \tag{0.4}$$

$$|\mathbf{b}_{31} \;\; \mathbf{a}_{31}| = \begin{vmatrix} 0 & 0 \\ k & 2 \end{vmatrix} = 0 \tag{0.5}$$

$$|\mathbf{b}_{12} \;\; \mathbf{a}_{12}| = \begin{vmatrix} k & 2 \\ 2 & 2 \end{vmatrix} = 2k - 4 \tag{0.6}$$

By (0.3):

$$\mathbf{b} \times \mathbf{a} = \begin{pmatrix} 0 \\ 0 \\ 2k - 4 \end{pmatrix} \tag{0.7}$$

Now, we find the norm:

$$\|\mathbf{b} \times \mathbf{a}\| = \sqrt{0^2 + 0^2 + (2k-4)^2} = |2k-4| \tag{0.8}$$

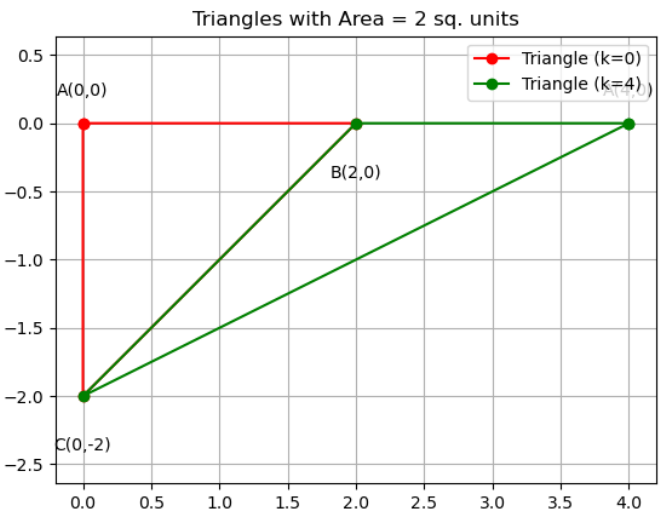
We solve for k:

$$|2k-4| = 4 \tag{0.9}$$

This gives two solutions:

$$2k - 4 = 4 \implies 2k = 8 \implies k = 4 \tag{0.10}$$
$$2k - 4 = -4 \implies 2k = 0 \implies k = 0 \tag{0.11}$$

The possible values of k are 0 or 4.



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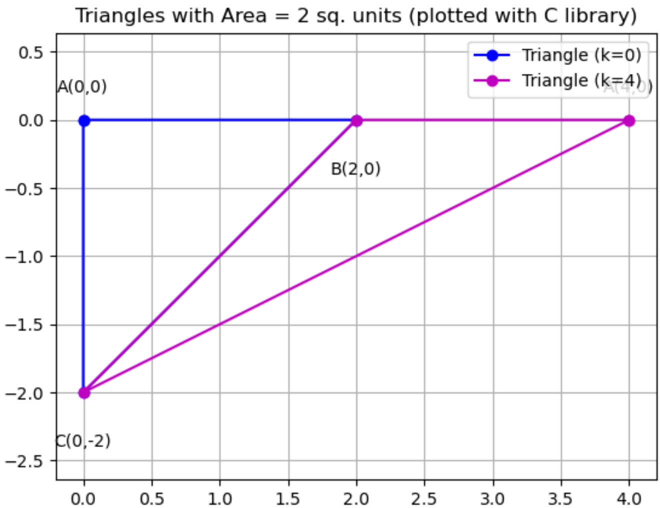
Triangles with Area = 2 sq. units (plotted with C library)
Triangle (k=0)
Triangle (k=4)
A(0,0)
B(2,0)
C(0,-2)
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

Fig